CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3^{rd} Semester of B. Tech. (EC) Examination

Nov-Dec 2012

EE-220 Electrical Machines and Power Circuits

Date: 01.12.2012, Saturday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Derive the EMF equation of a generator for simplex wave and lap wound generator. [07]

Q - 2 (a) Explain the significance of back EMF in case of DC motor. [04]

(b) Explain the characteristics of DC series motor. [05]

(c) In a long shunt compound generator, the terminal voltage is 230V when generator delivers 150A. Determine: (i) Induced EMF (ii) Total power generated (iii) Distribution of this power. Given that shunt field, series field, divertor and armature resistance are 92 Ω, 0.015 Ω, 0.03 Ω and 0.032 Ω respectively. [05]

OR

Q - 2 (a) What is necessity of starter in DC motor? Explain any one in detail with the help of necessary diagrams. [06]

(b) Derive EMF equation of single phase transformer. [04]

(c) A D.C. motor takes an armature current of 110A at 480V.The armature circuit resistance is 0.2 Ω. The machine has 6 Poles and armature is Lap connected with 864 conductors. The flux/Pole is 0.05 Wb. Calculate (i) Speed (ii) Gross torque developed by armature. [04]

Q - 3 (a) Derive the condition for maximum efficiency of single phase transformer. [04]

(b) A 230/230 V, 3 kVA transformer give the following results: [05]

O.C. test: 230 V, 2A, 100W

S.C. test: 15 V, 13A, 120W

Determine the regulation and efficiency at full load 0.80 p.f. lagging.

(c) State and explain the conditions for Parallel operation of transformers. [05]

OR

Q - 3 (a) Explain the production of rotating field in Induction motor with three phase supply. [08]

(b) Write short note on Auto transformer. [06]

SECTION – II

Q - 4	(a)	What is slip in induction motor? List out different methods for the measurement of slip. Explain stroboscopic method in detail.	[07]
Q - 5	(a)	Explain the torque slip characteristics of three phase induction motor.	[05]
	(b)	A 100- kW (output), 3300-V, 50-Hz, 3-Phase, Star-connected induction motor has a synchronous speed of 500 r.p.m. The full-load slip is 1.8% and full load power factor is 0.85. Stator copper loss = 2440 W. Iron loss = 3500 W. Rotational losses = 1200 W. Calculate: (i) the rotor copper loss (ii) the line current (iii) the full load efficiency.	[05]
	(c)	Why single phase induction motor is not self starting? Explain in detail.	[04]

OR

Q - 5	(a)	Explain double field revolving theory and derive the torque equation for single phase induction motor.	[07]
	(b)	Explain the basic principle of alternator and derive its EMF equation.	[07]
Q - 6	(a)	Explain the Synchronous Impedance method for determination of voltage regulation.	[07]
	(b)	Define substation. Give the classification of substation.	[07]

OR

Q - 6	(a)	What is tariff? Explain Two part,Three part and power factor tariff in detail.	[07]
	(b)	What is power factor? Give disadvantages and causes of low power factor.	[07]

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